Spatial representations emerge from temporal event credit in a hippocampus-inspired navigation agent without spatial supervision

Can spatial representations emerge from sensory inputs and learning objectives that do not explicitly encode space? We investigate this question using a bottom-up computational architecture inspired by hippocampal physiology, exploring whether spatial organization can arise from the temporal structure of sensory experience and self-generated behavior, without position labels, maps, or spatial rewards.

Our model combines dentate gyrus (DG)-like sparse sensory events with CA3-like sequential memory. Temporal relationships between encountered DG events provide two complementary learning signals: a **"push"** that reinforces DG activity more strongly after longer intervals between events, and a **"pull"** that rewards shorter transitions between sensory events or, in goal-conditioned variants, faster arrival at internally defined DG targets. Together, these pressures aim to make temporal separation a more informative proxy for navigational distance. The agent learns from egocentric visual experience and intrinsic motivations, without explicit information about its physical location or spatial distances.

During exploration of a continuous visual environment, spatially informative DG activity emerges. To isolate the contribution of temporal information, we compare temporally weighted DG feedback against mean-matched constant feedback, evaluating the resulting representations on identical sensory observations. Temporal feedback produces higher DG spatial information, less redundant response maps, and a broader spatial distribution of field peaks across five paired training seeds. These improvements extend to downstream decoder representations, where spatial information also increases consistently. In contrast, temporally weighting the policy's target-hit reward does not consistently improve exploration or spatial representation quality compared with constant hit rewards.

The agent's behavior further shapes spatial coding. Even with network weights frozen, the learned policy produces more spatially informative downstream activity than random actions, demonstrating that spatial coding depends not only on the learned network but also on how the agent interacts with its environment.

Together, our results demonstrate how spatial coding can emerge from the interaction of temporal learning, sequential memory, and self-generated behavior, without explicitly optimizing for spatial information.

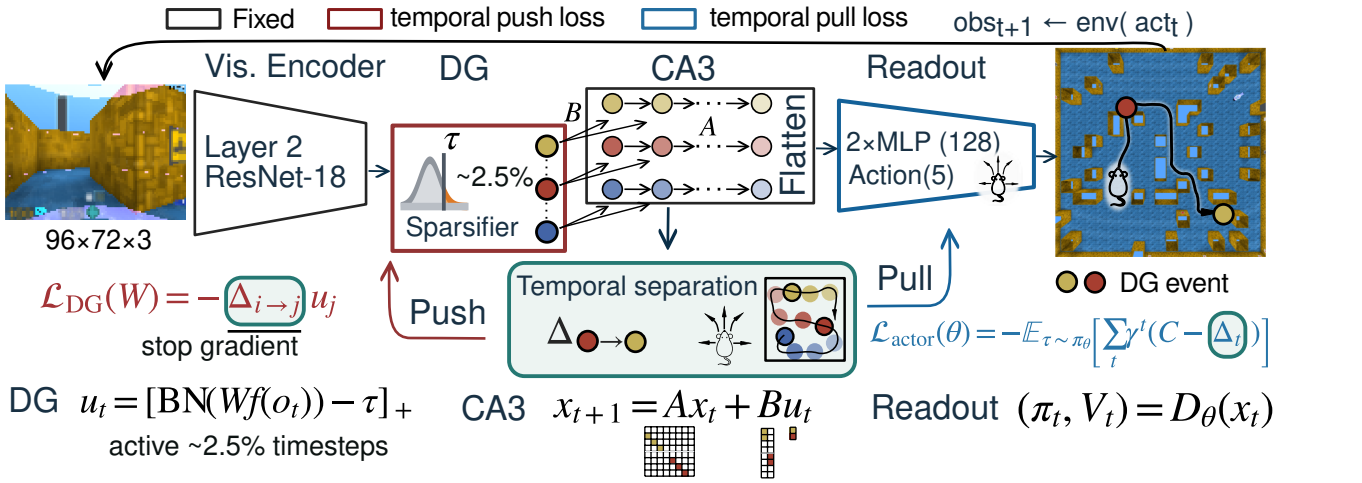


Figure 1: Push–pull architecture. A fixed pretrained visual trunk and trainable sparse dentate gyrus (DG) layer convert egocentric images into events. Fixed CA3-like memory stores event sequences; a learned readout controls actions and also receives egocentric depth. Red marks DG credit weighted by time since the previous eligible event (push). Blue marks policy reward for short event-to-event transitions or, in goal-conditioned agents, internal DG target hits (pull). C01 explores directly; C05 uses a DG target; C15 adds frontier-based target choice. No spatial coordinates, map, or spatial reward is supplied during learning.

Three tests. We measure spatial coding along evolving trajectories, change actions with network weights fixed, and replay identical observations through separately trained networks (Fig. 2). These comparisons distinguish the emergence of a spatial code during learning, its expression under different behavior, and the contribution of temporal DG credit to the learned network. Common replay removes evaluation-sampling differences, although training paths may still matter.

Spatial emergence and temporal push. DG coding rises online as the agent explores (Fig. 2A). With the same 10,001 observations replayed through each trained network, temporal DG credit improves the baseline DG spatial score in all five matched seeds (mean +0.0575), reduces map overlap, and broadens the area covered by eligible field peaks (mean +0.109; Fig. 2C,D). Seed 8 illustrates the largest peak-area increase among these pairs, while panel D shows the full distribution. Decoder-1 spatial information rises in all five baseline pairs (mean +0.680 bits per activation; Fig. 2E). The decoder effect also has positive means for both C05 worker arms, but one of the ten C05 pairs and one of the three C15 pairs reverse direction; the frontier-goal comparison remains limited to three matched seeds.

Behavior shapes spatial coding. With each network frozen, its learned policy elicits more spatially informative decoder-1 activity and more reliable spatial maps than random actions in all 38 matched models (Fig. 2B). Thus, behavior changes the spatial signal available from a fixed network. This test identifies an effect of the agent's interaction with the environment at evaluation time; it does not by itself show how policy changes shaped the network during training.

Temporal pull. Adding temporal weighting to the C05 policy's target-hit reward has mixed effects on visited coverage and online DG spatial information when encoder credit is held fixed (Fig. 2F). Both worker bonuses reward detector hits, and actor–critic discounting already favors earlier hits. A separate longer-credit intervention jointly raises PPO discount from $\gamma = 0.99$ to 0.999 and GAE λ from 0.95 to 0.99. Coverage AUC rises in all three prescribed-goal and all three C15 seed pairs, whereas frozen-rollout learned-unit spatial information falls in all six. Those frozen evaluations use separate policy-driven paths, so they do not isolate a fixed-input change in DG coding. Detector hits and coverage do not establish physical goal arrival; matched-command and physical-arrival analyses are needed to assess goal control.

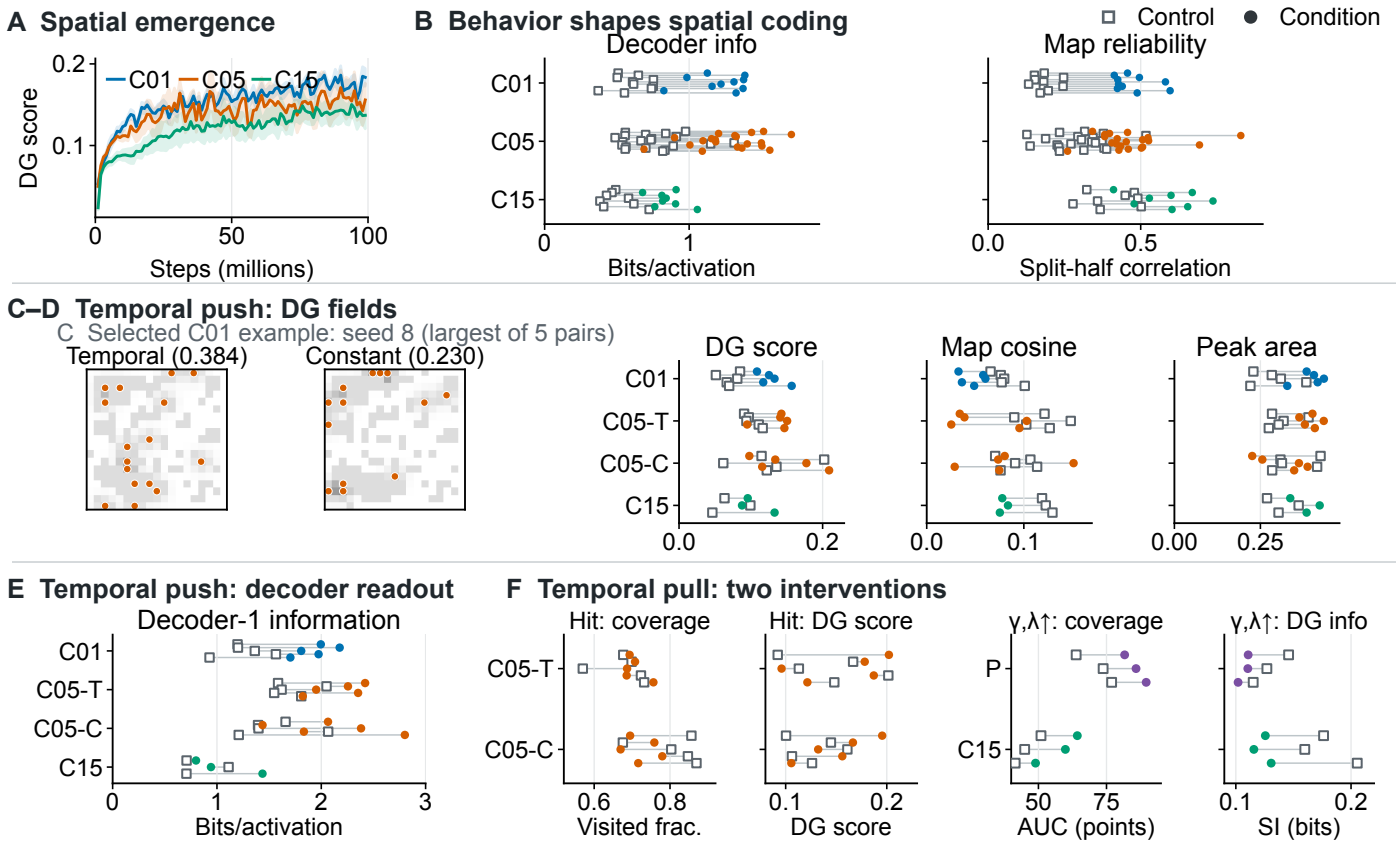


Figure 2: Temporal credit, behavior, and spatial coding. Blue, orange, green, and purple denote baseline (C01), goal-conditioned (C05), frontier-goal (C15), and prescribed-goal (P) agents. Open gray squares are controls, filled circles the tested condition, and gray lines connect matched seeds or frozen-model evaluations. The control and tested conditions differ by panel as stated below. **A,** Online amplitude-weighted DG spatial score on each policy's latest 10,000 observations: five-seed mean ±1 s.e.m. All curves use temporal DG credit; C05/C15 also use temporal worker hit credit. **B,** Frozen-model decoder-1 spatial information and split-half map correlation under random actions versus the learned policy. Each pair represents one of 38 terminal models, averaged over two 50,000-decision evaluations. Information is in bits per activation on equally weighted shared bins and jointly eligible units; correlation uses bins visited at least ten times in each episode-disjoint half of both policies. Both readouts favor the learned policy in all 38 models at this threshold. **C,** Baseline seed 8, selected for the largest peak-area difference among five paired seeds: temporal 0.384 versus constant 0.230 on the same 10,001 observations. Dots are DG-unit peaks after a 0.25% activity cutoff; grayscale shows Gaussian-smoothed peak density, and light gray marks unvisited bins. Effective peak area is exponential entropy of equal-mass peak density divided by 242 visited bins (100-unit Gaussian width). **D,** Temporal DG credit versus mean-matched constant control on the identical observation sequence at terminal checkpoints. Rows hold policy and worker bonus fixed: C01 ($n = 5$ seed pairs), C05 with temporal or constant hit bonus (C05-T/C; $n = 5$ each), and C15 ($n = 3$). DG score is amplitude-weighted; lower mean cosine among active-unit maps indicates less overlap; peak area is defined in C. **E,** Decoder-1 information per activation on the same replay, with temporal versus constant DG credit and the worker arm fixed. Temporal-minus-constant means are +0.680, +0.438, +0.559, and +0.217 bits per activation for C01, C05-T, C05-C, and C15; the numbers of positive pairs are 5/5, 5/5, 4/5, and 2/3, respectively. **F,** Two distinct tests of temporal pull. The left two plots compare C05 temporal versus mean-matched constant worker hit credit, with temporal (C05-T) or constant (C05-C) DG credit fixed ($n = 5$ pairs per row). Online visited-cell fraction and amplitude-weighted DG spatial score are averaged over 90–99 million training frames. Mean temporal-minus-constant effects are +0.026/+0.013 for C05-T and −0.087/+0.023 for C05-C (coverage/DG score). The right two plots compare longer PPO credit, which jointly raises γ from 0.99 to 0.999 and GAE λ from 0.95 to 0.99, against the original settings at 100 million frames. Coverage AUC rises by +14.2 points for P and +11.9 for C15 on average ($n = 3$ matched seeds each); mean learned-unit DG spatial information on separate 10,000-decision policy-driven frozen rollouts falls by 0.022 and 0.056 bits, respectively. The long-credit frozen paths differ between conditions, so this is not a fixed-input representation contrast. Coverage and detector hits are not executed goal control.